Devika Rani Sanaboyina

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Education

University of Maryland, Baltimore County — Aug 2024 – May 2026
***M.P.S. in Data Science**, GPA 3.86* | *Graduate Certificate: Project Management* — *Baltimore, MD*

Experience

Applied AI Engineer Intern — Jul 2025 – May 2026
Anthem Nation — *Remote, USA*
- Cut manual editing about **60%** and increased short-form output nearly **3x** with AI transcription, clip detection, and captions.
- Improved campaign readiness about **45%** and cut prep effort about **40%** with AWS workflows and CloudWatch monitoring.
- Built continuous S3-to-**Snowflake** ingestion with Snowpipe, Streams, and Tasks for KPI and agent-data reporting.

Data Analyst — Sep 2024 – May 2026
Retriever Essentials, University of Maryland, Baltimore County — *Baltimore, MD*
- Tracked campus food-support stockouts with a Python, SQL, and Tableau workflow for weekly inventory review.
- Modeled stockout risk with a lightweight classifier and threshold flags in a Tableau dashboard for weekly reordering.

Graduate Assistant, Data Analytics — Dec 2026 – May 2026
Office of the Dean of Students, University of Maryland, Baltimore County — *Baltimore, MD*
- Structured **9.6K** student address records into an analysis-ready dataset for consistent leadership reporting.
- Cut manual reporting time **60%** with a geospatial bubble-map dashboard of student location patterns.

Data Scientist — Aug 2025 – Dec 2025
Division of Student Affairs, University of Maryland, Baltimore County — *Baltimore, MD*
- Designed a bounded agentic AI assistant with LLM planning, IDF-weighted RAG, and Strands Agents.
- Built fail-closed routing for 5 crisis and policy intents, covered by **56 automated tests**.

Restorative Practices Summer Intern — Jun 2025 – Aug 2025
University of Maryland, Baltimore County — *Baltimore, MD*
- Structured multi-source university data to benchmark restorative practices and support strategic planning.
- Produced visual reports and coordinated workshop materials, scheduling, setup, and on-site support.

Data Scientist — Feb 2023 – Jul 2024
Orissa Power Consortium Limited (OPCL) — *Hyderabad, India*
- Supported a **6% relative rise in plant availability** (78.9% to 83.6%) through SCADA, generation, and outage analysis.
- Reduced **day-ahead generation MAE 18%** (34.7 to 28.4 MWh/day) with LightGBM using hydrology, weather, and seasonality
- Raised stoppage-detection recall from **46% to 84%** and median warning from **12 minutes to 6.5 hours** with LightGBM.

Projects

AgentShield – LLM Safety Evaluation Engine | *LangGraph, FastAPI, MCP, RAG, Gradio, SQLite*
- Built AgentShield to test **8 LLM risk categories/16 attacks** with LangGraph, RAG, MCP, FastAPI, and human review.
- Automated regression analysis: **8/8 aligned cases** moved FAIL→PASS with **0 regressions** across **90 passing tests**.

FairSlot: AI-Assisted Appointment Recovery | *XGBoost, OR-Tools, FastAPI, PostgreSQL, pgvector*
- Improved XGBoost calibration **28.3%** on **110,521 appointments** (Brier **0.1323→0.0948**) with **0.7653 ROC-AUC**.
- Advanced appointments **91 days** across **3,000 synthetic patients** with CP-SAT, FastAPI, PostgreSQL, and grounded RAG.

YouTube Trending Video Analysis | *Hadoop, HDFS, PySpark, Tableau*
- Processed a multi-GB, multi-region YouTube dataset through Hadoop, HDFS, and PySpark for engagement analysis.
- Built Tableau dashboards on views, likes, comments, and content-category trends across large-scale engagement data.

Hospital Performance and Resource Optimization | *Random Forest, XGBoost, K-Means, PCA, CRISP-DM*
- Random Forest reached R^2 **0.916** regression on **578K+** OSHPD records and **83.5%** classification accuracy.
- Clustered hospitals into **3 efficiency tiers** with K-Means and PCA for staffing, budget, and capacity planning.

Publications

- B. S. Kumar, V. A. Varma, **S. Devika Rani**, and A. Nishanth, "Fault detection and classification using machine learning," *Int. J. Res. Appl. Sci. Eng. Technol. (IJRASET)*, vol. 11, no. 4, pp. 3063–3069, Apr. 2023.

Technical Skills

Languages & Data: Python, SQL, PostgreSQL, SQLite, PySpark, Hadoop, FastAPI, REST APIs
ML & Modeling: PyTorch, scikit-learn, XGBoost, Random Forest, K-Means, PCA, TF-IDF, QLoRA
GenAI & Agents: LangGraph, LangChain, ChromaDB, MCP, Hugging Face, OpenAI Agents SDK, CrewAI, Strands Agents, Ollama
Cloud & Delivery: AWS (S3, Lambda, API Gateway, CloudWatch), Docker, GitHub Actions, Tableau, Power BI, OR-Tools